Innovation: The European Journal of Social Science Research

ISSN: 1351-1610 (Print) 1469-8412 (Online) Journal homepage: www.tandfonline.com/journals/ciej20

Introducing the Critical Making Responsibility framework for analyzing responsible innovation processes in grassroots practices

Regina Sipos & Maria Åkerman

To cite this article: Regina Sipos & Maria Åkerman (2023) Introducing the Critical Making Responsibility framework for analyzing responsible innovation processes in grassroots practices, Innovation: The European Journal of Social Science Research, 36:4, 652-672, DOI: 10.1080/13511610.2023.2195583

To link to this article: https://doi.org/10.1080/13511610.2023.2195583

Published online: 18 Apr 2023.

Innovation: The European Journal of Social Science Research, 2023
Vol. 36, No. 4, 652–672, https://doi.org/10.1080/13511610.2023.2195583

Introducing the Critical Making Responsibility framework for analyzing responsible innovation processes in grassroots practices

Regina Sipos[a*] and Maria Åkerman[b]

[a]*Institute of Vocational Education and Work Studies, Technical University of Berlin, Berlin, Germany;* [b]*VTT Technical Research Centre of Finland, Espoo, Finland*

(*Received 21 December 2021; final version received 19 February 2023*)

This paper introduces the Critical Making Responsibility Framework. The framework has been developed by the Critical Making consortium in order to analyze responsible innovation processes in grassroots innovation, specifically in the practice of making. The paper builds on a literature review to highlight the shortcomings of current Responsible Research and Innovation (RRI) frameworks' relevance towards grassroots innovation practices, and conversely, the lack of scientific understanding on ethics and responsibility in making. To fill the gap, this paper proposes a combination of the dimensions of the Grassroots Innovation Movements (GIM) analytical framework and the RRI capacity dimensions. Finally, the outlook reflects upon how the framework will be utilized in hands-on ways to support the work of academic and non-academic co-researchers of reflexive maker practices.

Keywords: literature review; critical making; grassroots innovation; Responsible Research and Innovation; participatory action research

Introduction

Academic interest in makerspaces has been growing, as these nurture an approach different from traditional methods of innovation: participatory grassroots innovation. Makerspaces are known for supporting the practice of making, which is a term for the subculture that constitutes a technology-based extension of DIY culture (Doyle 2013). This support consists of access to infrastructure and tools, but also to community members, their shared knowledge and skills.

Grand narratives around the possibilities the so-called maker movement represents have been formulated, and expectations towards these innovation spaces have been high, including bringing about a new industrial revolution (Anderson 2012) or democratizing innovation (Tanenbaum et al. 2013) through the prosumer's empowerment (Paltrinieri and Esposti 2013). Maker communities have also been celebrated for their potential impacts, including social (Unterfrauner et al. 2020; Bosse et al. 2019), political (Maxigas 2012) and environmental (Lange 2017; Kohtala 2016) impacts. A recent example of the capacity of the maker movement to address societal needs in a responsive manner was witnessed during the COVID-19 pandemic when grassroots communities were rapidly

*Corresponding author. Email: sipos@tu-berlin.de

Fan (2018) offers an interesting assessment of critical making pedagogy during the height of the pandemic

prototyping, testing, documenting and reproducing necessary products (Kieslinger et al. 2021) as global supply chains broke down.

Alongside the positive examples, academia has been debating whether the promises of positive societal, economic and environmental impacts are truly delivered. Moreover, some researchers have criticized the maker movement for technosolutionism and ideological colonialism (Lindtner, Bardzell, and Bardzell 2016). The American Make: magazine was criticized for their involvement with the US military (Finley 2012) or the MakerBot for becoming a closed-source project after it was built by a community of open hardware enthusiasts (Benchoff 2016).

similar issues have arisen with the Bambu 3D printing ecosystem

When discussing the making, it is important to note that it is not a uniform activity that follows one central blueprint and should be reproduced anywhere in the world exactly as prescribed. Following Ong and Collier's definition (Ong and Collier 2004) it is rather a 'global assemblage' (Lindtner, Bardzell, and Bardzell 2016) of hacker- and makerspaces, spaces of collaborative design and grassroots innovation, brought to life by offline and online communities that make use of the tools found in these spaces (Smith et al. 2017). Hacker- and makerspaces are localities that attract inquisitive people, early adopters and creators of innovative artifacts. Following from this heterogeneity, there are also different interpretations within the maker movement of the main societal goals of making and the meaning of some, supposedly key principles of maker movement, such as openness (Saari et al. 2021). However heterogenous, these spaces of grassroots technology innovation practices (Smith et al. 2017) provide researchers with opportunities to observe processes of collaborative, collective or community-based design (Bonvoisin et al. 2018). Makers and maker communities often engage in grassroots innovation, act as litmus tests to measure the (dis)contentedness of society, and pinpoint emerging societal needs and propose change (Sipos and Wenzelmann 2021).

As indicated by the response of makers to the Covid-19 pandemic, the potential of responsible innovation in the maker movement is interesting for research. The pandemic has shown the risks and limits of relying solely on global supply chains, centralized large corporations and governments, and therefore has set off an unprecedented interest in local, distributed, and openly accessible design and manufacturing to solve the massive unmet needs that have surfaced during the crisis. This disastrous period presented itself as a unique opportunity for makers to prove their skills. Such a shift, from niche to broader society impact, needs a critical view on social responsibility and ethics.

An apt example of what is revealed by moments of infrastructural rupture or breakdown

This paper suggests an integrated interdisciplinary approach to address responsibility in making which draws from the analytical framework proposed for research grassroots innovation movements (GIM, Smith et al. 2017) to understand the social embeddedness of maker movement and puts that into dialogue with the concept of Responsible Research and Innovation (RRI) capabilities developed within the academic debates on the responsibility of innovations (e.g. Stilgoe, Owen, and Macnaghten 2013).

Based on this dialogue, we introduce a first version of a Critical Making Responsibility Framework. To briefly summarize its background, the framework has been designed in the EU-funded *Critical Making* project to support the participatory research interventions of the project. The interventions aim at co-creating socially responsible and sustainable maker practices, increasing diversity and inclusion with regard to age and gender, and to support the idea of openness, while highlighting best practices both from the Global North (Germany, France) and the Global South (Brazil, South Sudan or Indonesia). The framework discussed in this paper will support this work to provide insight on the different dimensions of responsibility in the particular context of making, hereby illuminating the complexity of the topic. The authors believe that the analytical framework might support inquiry in other

Calling into question who actively designs the framework similar participatory interventions, especially those that allow for the cooperation between academic and non-academic co-researchers, e.g. in participatory action research.

A related academic discussion on critical making, which also influenced the framework presented here, stresses the advantage of combining reflexivity (and consequently also responsibility) with the particular opportunities that material practice, or making provides (Ratto and Hockema 2009). This approach inspired researchers to ask questions about communities that follow the underlying principles of responsible making, which are significantly less visible in the mainstream narrative of makerspaces compared to the corporatized version in which 'everyone just buys the kit' (Hertz 2012). The term critical making, coined by Matt Ratto, originally described participatory scholarly design practices that combine critical thinking and making (Hertz 2012). This unique practice, although it is a niche approach, is applied in academic work in ways that are relevant for the endeavor described in this paper, as critical making is extended towards real-world interventions. E.g. environmental research is transformed through citizen participation and technoscience (Wylie et al. 2014), interdisciplinary learning supports the combination of art, science and engineering studies with social interventions (Ratto and Hertz 2019) and Ratto's Critical Making Lab continues to extend the practice 'from scholastic humanistic work and into the realm of public intervention' (314, Ratto 2019) asking questions about reflexivity and practice. Moreover, the concept has also been used to describe a responsible and critical way of making as we can see in some forms of activism or GIM (Sipos and Wenzelmann 2021).[1] The Critical Making consortium refers to the concept of critical making for the first time as an approach for describing RRI practices in maker-innovator communities. Grounded in grassroots innovation processes, critical makers have been repetitiously highlighting values such as inclusion, ethics, responsibility, reflexivity, or openness (Sipos and Wenzelmann 2021). We find this discussion useful in identifying how critical thinking, which is an essential part of responsibility, is enacted and can be enacted in the existing practices of making.

While the framework is preliminarily aimed at supporting the work of the researchers within the project, in the future, it will be further developed to help researchers in participatory action research projects to better understand the criticality, reflexivity and responsible practices of innovation within grassroots, involving the points of view of grassroots practitioners. The current version presented in this paper is thus considered a starting point, aimed to inspire conversations and collect experiences from its practical application.

Analytical-conceptual background

Responsibility and criticality in making

The aim of the Critical Making consortium is to co-create a new responsibility framework that is particularly tailored to capture the responsibility issues that are relevant in making and grassroots innovations and respect their character. We know from observation and participation that grassroots innovators do engage in responsibility practices, as they advocate for openness, sharing, co-creation and user-centered and community-based design. The potential for responsible innovation inherent in a specific subset of grassroots innovators identified as maker communities is to be researched. This potential comes from the cumulative and shared knowledge of its members, and the technical possibilities that the democratization of innovation and access to technology through rapid prototyping offers, especially when combined with critical thinking and reflexivity. This is so far understudied and needs more research (Ratto and Boler 2014; Sipos and Wenzelmann 2021). However, we should not forget that innovating for the benefits of society is not the goal of a widely disseminated rhetoric of makers, who use standard engineering practices to solve issues with often limited societal impact. Furthermore, when we tried to find ready-to-use or easy-to-elaborate responsibility frameworks to be applied in the context of making, we realized that the recent debates of RRI largely neglect the questions of grassroots innovations, which means that they fail to address the particularities of citizen and community-driven innovations (see also Bhaduri and Talat 2020). In the following, we will describe how we attempt to overcome this failure by integrating the in-depth understanding of grassroots innovations studies (GIM) of how making is embedded in society and can provide a particular source of reflexivity with existing conceptualizations of responsible innovations developed in RRI literature. We start the description of our analytical-conceptual framework with RRI conceptualizations and then move on to introduce the GIM approach.

RRI capacities in grassroots innovations

The concept of RRI was developed a decade ago as a response to the need to deliberate the values embedded in the process of science and technology development and to provide a responsibility framework for research particularly in the context of European Union research and innovation funding (e.g. Gianni et al. 2019). It has been understood as primarily a tool to govern the socio-ethical aspects (Scholten and Blok 2015) of research and innovation and as such a continuation of broader debates in science and technology studies (STS) emphasizing the need to increase the sensitivity of researchers and scientific institutions on the needs, wishes and fears of citizens and communities (e.g. Irwin 1995). Particularly, there is a need to recognize that although science and technology development have enhanced well-being, they have also created unintended consequences, including climate change for example. In addition, many of the outcomes of science and technology development are controversial and have become matters of social concern as public discussions around issues such as genetically modified organisms and data economy show. Therefore, it has been claimed that a closer connection between science and society is needed (Gibbons 1999).

RRI as a concept is thus originally created to provide a responsibility approach for institutionalized research and refers to strategies and practices which enable researchers and a variety of stakeholders to become mutually responsive to each other and to anticipate the societal impacts of the outcomes of research. It is understood as 'a transparent, interactive process by which societal actors and innovators become mutually responsive to each other with a view on the acceptability, sustainability and societal desirability of the innovation process and its marketable products' (Von Schomberg 2011) What is to be noted is the importance put in the continuous engagement of citizens and civil society during the innovation process to negotiate the value of its outcomes. Following from that, RRI discussions are closely connected to and draw from the long-term tradition of STS studies in exploring means and practices of public engagement in science and innovations (e.g. Stilgoe, Lock, and Wilsdon 2014) and furthermore are also interested in finding ways to support the capacities of both citizens and other stakeholders and scientist and innovators to engage in fruitful communication (e.g. Selin et al. 2017; Tassone et al. 2018).

centrality of "community engagement" to RRI framework

In addition to putting emphasis of engaging stakeholders and public into dialogue with researchers and innovators, the RRI principles also highlight the need to integrate the practices of anticipatory governance into the innovation process (Felt 2018). According to (Stilgoe, Owen, and Macnaghten 2013), anticipation and forward-looking thinking together with collective stewardship are key factors of responsible innovation. Furthermore, they identify four particular RRI dimensions critical for responsibility in research: anticipation, reflexivity, inclusion and responsiveness, which according to Tassone et al. (2018) form the RRI capacity framework (see Figure 1.) that can be supported through education.

For the purposes of evaluation and monitoring the responsibility of research and innovation activities of research funding and performing organizations, RRI has been operationalized to six separate policy keys (gender, governance, ethics, stakeholder and public engagement, science education) and related indicators to be used in evaluation.[2] In addition, different RRI tools targeting the design and implementation of research process and to increase the sensitivity of research teams along the on-going work towards ethical issues and social responsibility have been developed.[3] As mentioned before, these tools have been so far mostly designed for institutionalized research and innovation. Following from that, it has been claimed that despite reflecting similar goals with both citizen and industry-driven pro-social innovation movements, the RRI discourse has remained mainly isolated from it. Furthermore, it has also been criticized of being fundamentally Euro-centric (Bhaduri and Talat 2020) designed for the societies and innovation systems of the Global North. Moreover, several scholars have claimed that the concept of RRI fails to recognize such conceptualizations of innovation which arise and are addressed towards the Global South including for example frugal and grassroots innovations (Mamidipudi and Frahm 2020). Such a distinction is relevant as the contexts in which those innovations are developed differ significantly from R&I practices of established research institutions and companies which means that the concept of RRI is not adoptable as such. As the sites of frugal and grassroots innovations are diverse, ranging from social collectives to informal enterprises, also the understanding of dynamics of innovations and their social embeddedness needs to be revised and enriched with context-specific knowledge. For example, Bhaduri and Talat (2020) claim that frugal innovations are by definition responsive to societal needs and values as they are directly

There are certainly notable gaps here in terms of developing a framework that is more specifically intended for community co-research, decolonial research, or community based maker and research practices

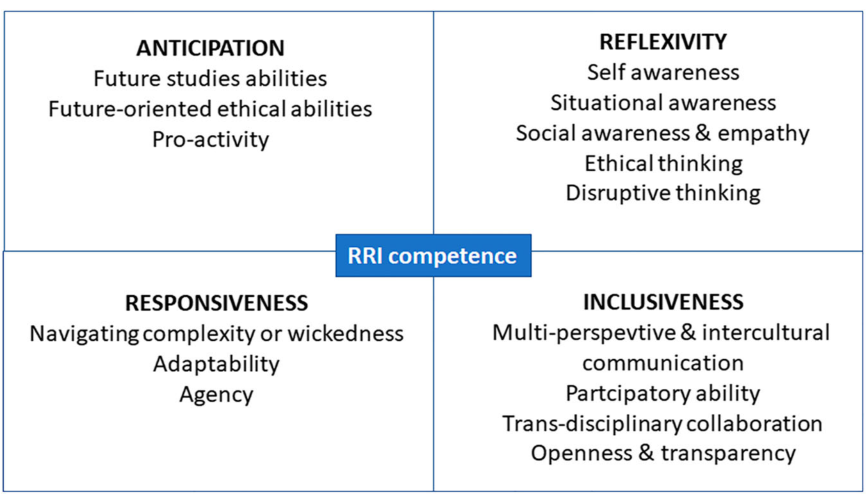

Figure 1. RRI competences. (Source: Tassone et al. 2018).

developed by the people and communities for the purposes of these same communities. In addition, the need for extensive foresight processes related to the potential outcomes of innovations is rarely needed as small-scale grassroots innovations mostly do not have broad social or ecological impacts.

Despite the criticism raised, and the failures of existing RRI frameworks to grasp the dynamics of grassroots innovations, the on-going debates on the definition, essence and different uses of RRI indicate that it is a flexible concept and framework open to interpretations and context-specific tailoring. We find it useful to explore how the four anticipatory and forward-looking capacities identified in the RRI literature can be used as resource in creating a responsibility framework for citizen and community-driven innovations. We believe that these principles can be adopted in a meaningful way for these purposes when enriched with an in-depth understanding of how the present state and future pathways of community-driven innovations are shaped by various societal processes. We introduce the GIM framework in the next section, which has been developed for exactly this purpose.

Grassroots innovation movements framework

Practical grassroots innovations committed to values of social justice or environmentally sustainable developments have existed for decades and caught the attention of researchers (see for example Schumacher 1973; Hess 2009; Smith 2005). The GIM framework that is guiding us during our work in this project is inspired by such previous work. The novelty of the GIM framework is that it proposes a multidisciplinary approach to provide researchers with a tool that sheds light on the particularities of grassroots from different viewpoints. Firstly, to analyze how GIM co-create alternative visions and practices of development, GIM uses insights of social movement literature on the mobilization of resources and political strategies for example. In addition, it draws analytical approaches from science and technology studies (STS) to direct the analytical gaze on the collective action frames for alternative science, (see Hess 2009) and from innovation studies to focus on concepts about knowledge creation, (see, e.g. decolonial scholars such as Escobar 2004), and technological innovation leading to alternative types of technological change. (Escobar 2004, 17–18).

pluriversality in design + maker practice

Making can be considered as an integral part of grass roots innovation movement. As Smith et al. summarize, 'we conceive hackerspaces, fablabs and makerspaces as a grassroots innovation movement because there is considerable activity outside formal institutions and because their networks are committed to exploring the social possibilities of bringing tools to people' (100, Smith et al. 2017).

Grassroots innovation communities, networks and movements create solutions to (hyper)local issues. This is mainly done in bottom-up, self-initiated processes, with a focus on and embedded in local communities and their problems, needs and values. Moreover, open-source documentation and sharing help solutions become more sustainable than universalist and closed-source approaches (Arancio 2021). Innovations are developed cooperatively, through collaborative deliberation, negotiation and while building trust. The solutions might be born out of necessity, to raise awareness around societal questions concerning marginalized groups of people, or, as Smith et al. point out, address issues neglected by conventional innovators, because developing such solutions is rarely profitable. Communities, discourse, knowledge, prototypes and innovations are emerging through the process, producing knowledge, appropriate technology and coordinating social organization (6, Smith et al. 2017).

Another relevant approach to grassroots innovation research which adds a layer of societal discourse to the technological discourse can be found in the anticipatory design of Human–Computer Interaction (Lindtner, Bardzell, and Bardzell 2016), an approach that resonates with RRI practices. This reflexive-interventionist approach has been used e.g. to study the aforementioned frugal and grassroots practices in the Global South, specifically to understand how so-called e.g. an Indonesian DIYbio collective contribute to citizen science by developing a water sampling protocol and a digital map of water data gathering, hereby challenging traditional beliefs about sites of technology innovation (Lindtner and Lin 2017). Research combining critical thinking, reflexivity or anticipatory practices with making, participatory design, or grassroots innovation is thus becoming more and more present in academia.

As mentioned earlier, Smith and his colleagues combined insights from the different disciplinary approaches and developed an analytical framework to better understand the pathways of development that different GIMs have walked. The framework is designed for researchers to analyze already established GIM retrospectively, i.e. to gain better understanding of the pathways developed autonomously over time, without the intervention of researchers. Smith et al. (2017) argue that pathways to sustainable development are plural, and they want to know more about how groups and networks address questions of development, how they express relevant values in the innovation activity and what shapes their pathway through that activity. The authors also claim that broader social visions and implications of developments are made richer by analyzing GIM (Smith et al. 2017, 5): it is a matter for analysis to understand how GIM provide 'a source of reflexivity in society, by pointing to the contention and plurality involved in sustainable developments and opening up more spaces for doing the politics of alternative sustainabilities'. It is this question of reflexivity within the communities and its extension that becomes a source of reflexivity in society that makes this endeavor interesting for RRI. In our Critical Making Responsibility Framework, we aim to strengthen the potential of this reflexivity by combining the retrospective analytical framework of GIM with the forward-looking RRI capacity framework.

The GIM framework suggests analyzing 4 interrelated concepts to understand GIM: broader contexts, framings, spaces and strategies and pathways (see Figure 2). We use these four dimensions in the Critical Making Responsibility Framework to identify which factors and processes are important when targeting the particularities of social responsibility and RRI in making.

Critical Making Responsibility framework

Method for developing the framework

In parallel with scholars, grassroots practitioners themselves also started developing strategies to help others achieve more sustainability when developing new technologies in makerspaces. One example is the set of 'Sustainable Making Principles' (Nuesse and Wanalo 2019). The Critical Making consortium has tailored a working definition for responsibility in making and grassroots innovations that guides the development of the Critical Making Responsibility framework. This is based on three sources: first, the aforementioned set of strategies, which was co-developed by a consortium member and grassroots practitioners, second, the RRI and GIM literature reviews and third, insight provided by critical making scholarship.

The consortium's working definition is as follows: Responsible innovation and making in grassroots practices means that those who tinker with existing technologies

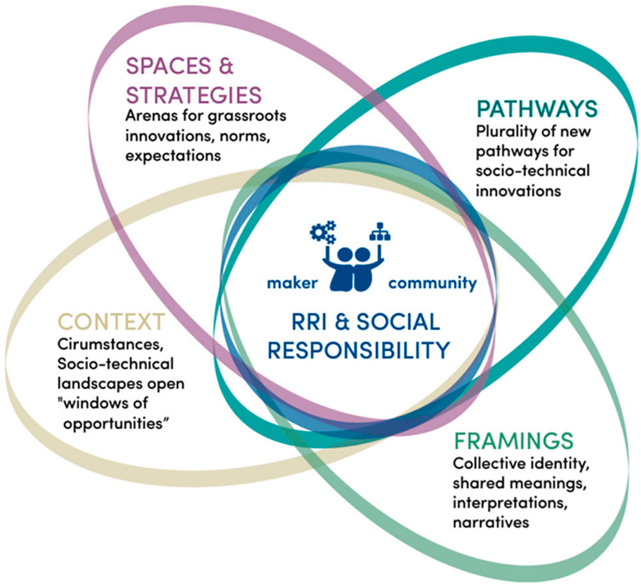

Figure 2. Critical Making Responsibility framework. Illustration by the Critical Making consortium.

and develop new solutions do this critically. This criticality reflects on their responsibility to ensure that the solutions are locally situated and community-based; responsible, ethically correct and addressing societal needs; based on critical thoughts and reflections about power structures; aim to have positive impact and change structures; and the practice itself is joyful and meaningful to the makers.[4]

To further develop these tailored approaches to responsibility, the Critical Making consortium started to create a dialogue between the chosen two analytical frameworks, i.e. the RRI capacities and the GIM framework during the first phase of the Critical Making project from January to June 2021. The first step was to create a matrix that explores how the forward-looking and reflexive RRI principles and retrospective GIM framework enrich each other. We have particularly wanted to understand and explore which dimensions align with each other and which might not.

An important initial step has been the alignment of different disciplinary terminologies to a common understanding as we intend to combine concepts stemming from distinct disciplinary backgrounds. Previous experiences have shown that terminology is crucial when defining a cross-disciplinary approach. While terms and concepts driven from the GIM and RRI frameworks like ‘context’ and ‘framings’ or ‘anticipation’ and ‘reflexivity’ might be termed researchers of a particular discipline immediately understand, such terminology is not necessarily accessible for researchers of other disciplines and other non-academic actors who might be interested in using the methodological tools developed in the project.

A first 4×4 matrix has been created by the authors of this manuscript in an attempt to combine each of the dimensions, based on thorough literature review. Eight multidisciplinary researchers with backgrounds in social sciences and STS, experience with maker communities, GIM and/or RRI, as well as disciplinary foci on gender, education and specializations in open source technology and development shared experiences. They explored the feasibility and opportunities of developing the matrix and engaged in discussions on its usefulness and practicalities of implementation. The aim of these reflexive exchanges has been to create a preliminary framework of short descriptions of anonymized examples from our research experience[5], which will support the understanding and the thought process of the future users of the framework. The exchanges also helped identify elements which we currently see as not applicable.

RRI meets GIM

As described in the previous section, the definition of the four RRI capacities is developed for institutionalized actors engaged in funding and performing research and innovation activities and basically for the socio-technical systems of the Global North. Therefore, although we take the these RRI dimensions as a starting point to develop the Critical Making Responsibility Framework, we need to reconsider the meanings of the dimensions when applied for making and grassroots innovations. For example, as mentioned earlier, according to (Bhaduri and Talat 2020) grassroots innovations are usually incremental with little risks which means that broad foresight processes are not relevant and often too heavy in these contexts; although anticipation is still a meaningful activity. We must also take into account that grassroots innovation – especially in the Global South – happens with very limited resources, and with hacked, tinkered, pirated technologies but in return (and opposed to conventional innovation) with a very high involvement of local communities and everyday people in the co-design process. In addition, also reflexive processes need to be designed in different ways and question of inclusiveness gets different meanings at the grassroots level compared to institutional research contexts. For this adoption, we have put the RRI dimensions into dialogue with the GIM conceptualizations and created a 4 × 4 dimensional analytical framework (see Table 1).

Elaborating the axes of the framework

Vertical axis: the four GIM dimensions

The **context** helps outline the conditions in which the movement is developing. Historical, political, economic, cultural, religious contexts that could be generative or constraining, and other circumstances, issues and situations, including opportunities available within those contexts that had a generative effect on the movement are considered here.

In **framings**, future possibilities are negotiated collectively, including establishment of shared vision(s). Framing is the process of meaning production that helps communities connect to powerful narratives beyond shared grievances which can be expressed in critique towards mainstream practices. Framings are shaped by underlying assumptions, and can include problems, strategies, requirements, theories, knowledge, design criteria, exemplary artifacts, testing procedures and user practices that emerge through social interaction. While it can include technological frames (free/open source software, free/open source hardware, peer production, personalized manufacturing, mass customization and a new industrial revolution, the democratizing power of technological citizenship), it can also include or exclude a broader set of framings, such as social, economic, or political questions and can be important factors in how they design their practices.

Spaces and strategies crystallize novel strategies and co-operative forms. What actions communities take, and how those actions are influenced by the availability of resources is explored here, considering that spaces can not only be physical, but also social, discursive and institutional (makerspaces are spaces for grassroots digital

Table 1. GIM-RRI matrix: the Critical Making Responsibility framework. Table created by the Critical Making consortium.

		RRI competence			
		Anticipation	Reflexivity	Inclusiveness	Responsiveness
GIM Dimension	Context	Ability to understand and act upon the ongoing changes in social, historical, political, economic, cultural, religious contexts (trends & weak signals) and other circumstances and what kind of opportunities, restrictions and requirements they may provide in the future.	To become aware of how social, historical, political, economic, cultural and religious context have affected on ones activities (innovations, projects etc.) and what kinds of contexts their reactions & innovations might create, (e.g. vicious circles or hope, and for whom?)	To become aware of exclusive, contextual patterns - to understand that you don't by accident exclude others (like women, elderly, etc) - understanding how exclusion works and supporting people based on the contextual patterns of exclusion	To understand the particular societal needs arising from the context and to respond to them through making & innovations and in addition knowing 'how to react and whom to contact to influence the societal rules of the game'.
	Framings	not applicable	To become aware of how used language and terminology shapes the taken actions and what kinds of values and interests are mobilized, maintained or challenged with the language used. Shared framings can help and hinder dialogues and once that is recognized, something new can be learned.	To reflect upon and become aware of the wordings that are used, or the setup of the space, and whether they create inclusion or exclusion? Does the shared umbrella of interpretation lead to missing any perspectives?	not applicable
	Spaces/ Strategies	To become aware of one's own strategies to act, to learn to deliberately build strategies towards desired futures and to be able to anticipate what kinds of futures (and future spaces of action) the applied strategies create.	To become aware of how chosen strategies influence other people or environment - what are the risks and rewards for the surrounding community and environment of the chosen strategies	To become aware of the norms and conventions that 'made the space' of making & innovations: if excludes someone, become aware of these norms and conventions, physical structures and language.	To explore how available resources will influence what you do (skills in the team; tools available) and how to act to expand them.

(*Continued*)

Table 1. Continued.

	RRI competence			
	Anticipation	Reflexivity	Inclusiveness	Responsiveness
Pathways	To become more aware of what sort of pathways are supported: what future pathways are made while doing concrete projects, and reflect upon the potential plurality of it, to anticipate the impact of the ethical pathways. To recognize the path dependencies, become aware of what one can change with the created pathway and what not.	To become aware of one's own role and the situatedness of the activities carried out: how those impact/influence the environment. By recognizing the various pathways (anticipation), the potential social and ecological impacts can be reflected upon.	To reflect upon whether the developed or imagined pathways maintain existing exclusive structures, do they create new exclusions, new divisions between people? How can they be made more inclusive?	To investigate what kind of support the desired pathways would need in the broader social context (knowledge, funding, policy changes etc.) and/ or whether they may face resistance and to consider how this support can be gained and resistance addressed.

fabrication, maker movements and grassroots groups, activities include educational outreach and skills provision, etc.). Locations and activities that enable them to do experimentation and innovation differently are analyzed, actions done by enrolling audiences, alliances and users to improve their own performance (in a user-centered way, creating public engagement) and making alternative spaces of engagement. It is hereby that resources are mobilized while grassroots consider the costs and benefits, risks and rewards of strategies, shaped by the conditions attached by resource holders that influence the outcome of their activities (Smith et al. 2017, 26).

In the **pathways** section, various opportunity pathways are constructed and assessed from multiple perspectives. How does the plurality of pathways contribute to alternative developments over time? Ideas and aims are continuously developed and dismissed; objects and practices and their materiality also contribute to developments in different and changing settings over time, including a future perspective. These alternative pathways and their plurality show that there is not just one self-evidently best pathway, and the political nature of grassroots movements might contribute to new pathways created with greater attention to issues of social inclusion, diversity and difference and social justice, playing a key role in their RRI practices (Smith et al. 2017, 28).

Horizontal axis: the four RRI procedural responsibility dimensions

Anticipation refers to systematic thinking which aims to increase resilience of communities and helps to recognize and create opportunities for challenging the existing state of the art with novel social and technical innovations. Anticipation can be fostered with various participatory and deliberative foresight tools including horizon scanning, scenario building and road mapping. The aim is also to make people aware of existing social imaginaries.

Reflexivity refers to deliberate rethinking of how one's activities encounter and reflect the social norms and conventions and potentially challenge or strengthen existing social power relations, division of labor and costs and benefits or whether it causes potential risks for other people or ecological environment. Reflexivity is a process of questioning one's own activities and looking at them from the perspective of other people and natural beings.

Inclusiveness, which is the third RRI dimension refers to the need to include multiple voices and stakeholders in the innovation and making to bring in legitimacy and to provide an opportunity for stakeholders to express their concerns and opinions about the direction of activities. Several engagement methods to achieve inclusion in research have been introduced including for example citizen juries and panels or more light consultation through surveys and polls. In grassroots innovations the context is different as innovations are driven by citizens. In this case also, there is the need to carefully consider that people with multiple background feel welcome and get their voices heard in making activities and to make sure that also often underrepresented citizens (e.g. elderly people, young people, people with lower socio-economic status, etc.) are invited to participate.

Responsiveness is the ultimate aim of the three previous RRI principles: to increase the capacity of researchers and science and innovation system to be responsive for social challenges related to their research. In institutionalized research, this kind of responsiveness is shown for example in the direction of research efforts towards recognized societal challenges. In addition, research actors can actively influence the rules of the game in society by promoting changes in regulation and standards and contributing to on-going policy debates and programs.

Prompts for using the framework

To reiterate, we understand that not all researchers, and especially non-academic co-researchers in participatory projects might be familiar with all the terms used in the matrix. To support their practice-based work without overburdening them with additional readings, the Critical Making consortium made a first attempt in translating academic terms into a short, succinct description and real-world examples that might bring those high-level ideas closer to their practice. Below follows the matrix converted into a list of accessible prompts which are informed by the field research experiences of the Critical Making consortium.

CONTEXT

Context × Anticipation

- Explanation: Anticipation here is the ability to understand and act upon the ongoing changes in social, historical, political, economic, cultural and religious contexts (including trends & weak signals) and other circumstances. It also refers to what kind of opportunities, restrictions and requirements these may provide in the future.
- Example: One could explore community-based innovation processes that reflect upcoming societal changes: grassroots innovators being first sensitive to societal change and reacting by kickstarting innovation, because innovative capabilities are based in community. Viewing trends in making from the industry's point of view, the spread of makerspaces could be a sign of distributed manufacturing becoming more prevalent.

Context × Reflexivity

- Explanation: Reflexivity refers to becoming aware of how social, historical, political, economic, cultural and religious contexts have affected one's activities (including innovations, projects, programs) and what kinds of contexts their reactions and innovations might cause (e.g. vicious circles or hope, and for whom?)
- Example: While designing a participatory project, a responsible researcher or maker needs to ensure that visibility does not cause harm to its participants, for example in projects that tackle human rights issues or might generate knowledge uncomfortable for decision-makers. A case of this was the negative, unintended impact in a grassroots innovation project trying to help homeless people by developing water filtration tools, but as newspapers started reporting about them, people in the settlements, who were considered illegal, got evicted.

Context × Inclusiveness

- Explanation: To become aware of exclusive, contextual patterns. It is necessary to understand these in order to not (even if by 'accident') exclude others. This is especially applicable to women, elderly, and other, traditionally underrepresented groups. It is crucial to understand how exclusion works and support people based on the contextual patterns of exclusion.
- Example: Projects proactively designed to include underrepresented communities and develop frameworks that support their inclusion based on the context. An example is a capacity-building project that develops the self-esteem of minorities

and allows them to become part of a 'time share bank' for participating in incubation programs, instead of having them pay, thus, building an alternative economy.

Context × Responsiveness

- Explanation: To understand the particular societal needs arising from the context and to respond to them through making and other types of innovations. In addition to this, knowing how to react and whom to contact to address the societal needs and risks related to novel innovations or identified during making.
- Example: Responsive makers and grassroots innovators are those who directly address the needs of community. Responsiveness could also mean having the networks and ability to reach e.g. local politicians to generate influence on higher levels and achieving the goal through policy change or other types of support.

FRAMINGS

Framings × Anticipation

- Anticipation relates to forward-looking activities and deliberate actions aiming to affect future pathways whereas framing as an academic term refers to existing shared meanings and cultural structures that shape these meanings. A small group of actors often only has an impact on broader cultural discourses and assumptions once the community has grown into a movement. In addition, it is also difficult to anticipate the changes in these structures within which they need to carry out their work.

Framings × Reflexivity

- Explanation: To become aware of how the language and terminology used shapes the actions taken, and what kinds of values and interests are mobilized, maintained or challenged with the language used. Shared framings can help and hinder dialogues. Once this is recognized, a learning process can begin and change might occur.
- Examples: Reflecting upon the framings we work with might reveal how different people understand the terms free, open source, open innovation and how different community members' experiences might clash in these wordings. Framings of different concepts, such as nationalist, leftist/socialist or capitalist framings of social innovation are influenced by the country where it takes place and its history. Another example could be framings of beneficiaries in fundraising processes: for this purpose, they are often described as passive, 'in need of help', downplaying their abilities to contrast with the abilities of those who will be funded to deliver that necessary help.

Framings × Inclusiveness

- Explanation: To reflect upon and become aware of the wordings that are used in verbal or written communication, or the setup of the space one creates for the community. Does a specific setup lead to inclusion or exclusion? Does the shared umbrella of interpretation within the existing community lead to missing any perspectives?
- Examples: Creating shared interpretations is necessarily a collective, discussion-based process. When a member of the community argues for a particular idea,

other perspectives are automatically downplayed. This collective production of ideas and meanings creates bonds but might also exclude others. Does the term 'maker' exclude 'makeuses' and vice versa?

- Framings × Responsiveness
- Not applicable: Similarly to the Anticipation × Framings, we find that the intersection of Framings × Responsiveness is not an applicable category. The reason is that framings cannot be influenced through policies, standards and public action but are rather changed slowly over time, through collective reflection.

SPACES & STRATEGIES

Spaces/Strategies × Anticipation

- Explanation: To become aware of one's own strategies to act, to learn to deliberately build strategies towards desired futures and to be able to anticipate what kinds of futures (and future spaces of action) the applied strategies create.
- Examples: Strategies are always forward-looking in themselves, with an explicitly or automatically embedded idea of which directions to take and why. By asking the participants what their goal is for the next years, or what kind of world do they want to see then and how does their project help them reach this, such strategies of anticipation can be mapped.

Spaces/Strategies × Reflexivity

- Explanation: To become aware of how chosen spaces and strategies influence other people, including what the risks and rewards for the surrounding community and environment of the chosen strategies are. After deliberating the strategy itself (as it might be something that was not consciously planned), one might ask themselves: What are then the 'side effects' of the strategies communities have chosen?
- Examples: By saying no to taking money from a big company, an already underfunded community remains low on financial resources, however, their practice stays uninfluenced. Instead, they decide to use limited but non-attached resources to avoid outside powers impact their values and practices in ways they deem as negative. Another example is when a community receives particular machines free of charge. If this is a 3D printer, they might move away from paper prototyping and create more plastic waste than previously in the process, which becomes an unintended impact caused by the resources they have.

Spaces/Strategies × Inclusiveness

- Explanation: To become aware of the norms and conventions that 'made the space' in terms of making or grassroots innovations. What has contributed to it including particular people, and if someone is excluded, there is a need to become aware of those norms and conventions, physical structures and language that contributed to the exclusion. There is a need to become aware of what capabilities and skills are expected from people to be allowed to participate.
- Examples: In addition to physical inclusiveness (accessibility or safety of space, tools, website), cultural, and other influences might also play a role. In some countries, cultural issues might play a role, such as it being inappropriate for women to leave their homes in the evening. This has led to only men meeting in

the spaces created for the whole community in a project. The issue was reflected upon, and additional activities were planned from then on during the daytime hours.

Spaces/Strategies × Responsiveness

- Explanation: To explore how available resources will influence what you do and if the resources or chosen strategies limit the scope of social goals you address? How to act to expand the resources and whom to engage in commenting and reflecting the chosen strategies?
- Examples: It might be explored what skills are available within the team and which tools they have access to. Was there a case when they wanted to do something but their skills, tools, space, resources didn't let them, so they pivoted and did it differently? Did this modification still develop a suitable solution? How was this possible?

PATHWAYS

Pathways × Anticipation

- Explanation: To become more aware of what sort of pathways are supported. What future pathways are consciously or subconsciously made while doing concrete projects? These need to be reflected upon, including their potential plurality to anticipate the impact of the ethical pathways. The goal is to recognize the path dependencies, become aware of what one can change with the created pathway and what not, for example through envisioning: what is the future the project is aiming at, and what are the different pathways to get there?
- Examples: Acknowledging that prerequisites need to be achieved before efficient change is done is crucial. While change might be blocked by existing structures, but with long-term planning of a pathway, one can have an impact. An example of such long-term planning of hidden agendas includes community network projects that at first glance are about physical infrastructures, however, their ultimate goal is empowering and protection of the rights of indigenous communities.

Pathways × Reflexivity

- Explanation: To become aware of one's own role and the situatedness of the activities carried out, including how those impact/influence the environment. By recognizing the various pathways (anticipation), the potential social and ecological impacts can be reflected upon.
- Examples: If a maker community decides to opt for distributed manufacturing, they ought to recognize their own role in making various pathways happen. These pathways can be based on business and start-up culture, or can be more environmentally or socially just, representing changes the maker movement significantly contributed/can significantly contribute to.

Pathways × Inclusiveness

- Explanation: To reflect upon whether the developed or imagined pathways maintain existing exclusive structures, do they create new exclusions, new divisions between people? How can they be made more inclusive?

- Examples: The long-term work of a social innovator and activist lobbying for internet laws to be more open in the late 1990s to turn his country into a knowledge-based society combined with the completely separate work of another social innovator bringing micro-hydro plants for sustainable electricity to remote areas combined enable remote communities today to have their own community-maintained electricity and internet without being hindered to do so by the market or complicated laws.

Pathways × Responsiveness

- Explanation: To investigate which societal actors and resources need to be engaged to support the realization of desired future pathways and whether there is a need of policy or regulatory changes.
- Example: For open hardware in healthcare, a project has explored pathways which, depending on the cultural context, needed legal changes, e.g. in the medical device legislation, to be adapted in order to become available. Makers might strategically engage with academia to receive new ideas, cooperate on interdisciplinary projects, or visibility through scientific articles, leading to more impact to change future pathways. Some also engage with governments on different levels, or the United Nations, not only to receive funding, but also to influence e.g. the political support of creative economy and so shape a desired future pathway.

The benefits of cross-fertilizing GIM & RRI frameworks

The cross-fertilization of the GIM and RRI frameworks benefits both discussions. The analytical conceptualizations of GIM are originally developed for researchers to understand the dynamics of social movements. In that sense, it provides an outsider's perspective to grassroots innovations. Currently, it is not used to support grassroots processes or as part of participatory action research. While combining it with the RRI capacities, we want to elaborate this framework towards a reflexive tool that is useful for both researchers interested in multiple dimensions of social responsibility in grassroots innovation but also for practitioners to support their reflexive processes. Following that the combining of the general RRI principles with the GIM insight on grassroots innovations as a social movement will also enrich both theories. For the RRI discussion, the dialogue with GIM broadens the concept of innovation which has so far been restricted to institutional innovation settings. For the GIM community, it provides new approaches for future-oriented participatory action research.

Summary and outlook

Although the Critical Making Responsibility Framework has been developed only recently and still needs to prove its validity in practical and analytical contexts, first feedback from actors in the field is very encouraging.

On the one hand, the Critical Making consortium has already made use of the framework to create awareness about criticality and responsibility in maker settings and the framework has contributed to shaping the co-design of interventions. The outcomes of these participatory and co-creative sessions were summarized in parallel with the peer review process of this paper and were shared at the FAB17 conference to inform wider audiences about those first experiences in applying the framework, including concrete questions developed by the co-designers based on the Framework and pertinent to the cases (Sipos et al. 2022a).

In the next phase, the consortium will further validate and elaborate the descriptions within the case actions, and then with field actors to get their insights on the relevancy of suggested responsibility dimensions. Once the general framework is ready, we move on to develop specific toolsets (including concrete guiding questions) for:

1. researchers that engage makers in participatory action and for
2. grassroots innovators in the field (e.g. makers).

This is done to create a self-reflective tool for people and communities who ask themselves whether their practice is sustainable, responsible, etc. and also to provide a tool for researchers and policymakers who want to assess the responsibility of particular actions. The Critical Making Responsibility Framework is thus to be further developed into hands-on tools, tested and iteratively developed in a variety of co-designed case actions within the Critical Making project.

While developing a questionnaire for self-evaluation and planning is relatively straightforward, it is not necessarily fun or engaging. Overburdening people with seemingly theoretical or rhetorical questions who would rather be tinkering cannot lead to useful outcomes. Thus, we will explore the possibility of accessible, easy-to-use, hands-on, visual tools that allow the process to be gamified, with reflexive questions as prompts. The medium will be decided upon together with the practitioners and tested and iterated together with them, and the language will go through further 'translation' processes to make the tool accessible in non-academic language and to be made hands-on and practical to be used before (for planning), during (for reflection) and after (for self-evaluation) any grassroots community and project. As maker communities often work, develop and think together, grassroots making is a non-linear and very creative process. We are curious to see: while grassroots often act in very hands-on ways, what happens if maker communities are given or give themselves the tools, time and permission to think, anticipate, reflect together?

Notes

1. A more detailed literature review of the term can be found in the Critical Making Baseline report, developed to support the work of the Consortium (Sipos, Åkerman, and Wenzelmann 2022b). This report includes the analysis of various relevant practices, including its roots in critical technical practice, critical design, adversarial design, critical engineering, tactical media, etc.
2. See for example MoRRi or SuperMoRRI.
3. See for example the RRI-tools portal https://rri-tools.eu/ or the NewHoRRIzon societal readiness thinking tool https://newhorrizon.eu/thinking-tool/.
4. The working definition has been outlined in the Critical Making Baseline Report, which is yet to be published. At the same time, to "give back" to makers, a methodological toolbox translating the principles back to the everyday practice is being developed and tested in a participatory manner. See https://zenodo.org/record/5948298#.YoIx8C-23BI
5. The cases are anonymized to protect the identities of those who could be negatively affected by having their names or projects revealed, as they work in complicated political contexts. However, as a reviewer pointed out, referencing the projects adds relevance to their theoretization. The project highlights best practice cases – whenever safe and appropriate – on the https://criticalmaking.eu website and on social media.

Acknowledgements

We would like to thank Barbara Kieslinger for her thorough revisions and continued support, Hanna Saari and Christian Voigt for reviewing the Critical Making Baseline report, which included the first draft of this framework, as well as the Consortium members for their contribution to its development.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This project has received funding from the European Union's Horizon 2020 Research and Innovation programme under grant agreement No. 101006285.

Notes on contributors

Regina Sipos is a Research Associate at the Technical University of Berlin, focusing on postcolonial computing, critical technical practice, collaborative technology design and social innovation in grassroots communities. She has around 15 years of experience working with grassroots innovators and building participatory and open platforms for co-creation, and held multiple executive-level, advisory board and steering committee roles supporting citizen science, social innovation and grassroots innovation communities. Successful third-party funded projects she co-developed include Critical Making and Infrastructuring in Grassroots Innovation.

Maria Åkerman acts as a Principal Scientist at VTT Technical Research Centre of Finland. She has multidisciplinary social scientific background in environmental policy, science and technology studies and economics and her key research interests include socio-technical transitions, sustainability governance, politics of environmental knowledge production and citizen and community-driven innovations. Working at the interface and across various disciplines, she has also been interested in methodologies of interdisciplinary research and collaborative and actionable knowledge production. Her recent projects have focused particularly on circular economy transition, energy transition and citizen engagement in climate action.

References

Anderson, C. 2012. *Makers: The New Industrial Revolution*. New York: Crown.

Arancio, J. C. 2021. "Opening Up the Tools for Doing Science: The Case of the Global Open Science Hardware Movement." *International Journal of Engineering, Social Justice, and Peace* 8 (2): 1–27. doi:10.24908/ijesjp.v8i2.13997.

Benchoff, B. 2016. "The MakerBot Obituary." Hackaday.Com. https://hackaday.com/2016/04/28/the-makerbot-obituary/.

Bhaduri, S., and N. Talat. 2020. "RRI Beyond Its Comfort Zone: Initiating a Dialogue with Frugal Innovation by 'the Vulnerable'." *Science, Technology & Society* 25 (2): 273–290. doi:10.1177/0971721820902967

Bonvoisin, J., T. Buchert, M. Preidel, and R. G. Stark. 2018. "How Participative Is Open Source Hardware? Insights from Online Repository Mining." *Design Science* 4: 1–31. doi:10.1017/dsj.2018.15.

Bosse, I., D. Krüger, H. Linke, et al. 2019. "The Maker Movement's Potential for an Inclusive Society." *Atlas of Social Innovation. 2nd Volume: A World of New Practices* –: 201–206.

Doyle, C. 2013. "Making Makers." In Ashling Hotel, Dublin, Ireland. http://www.lin.ie/conference/conference-2013/.

Escobar, A. 2004. "Beyond the Third World: Imperial Globality, Global Coloniality and Anti-Globalisation Social Movements." *Third World Quarterly* 25 (1)): 207–230. doi:10.1080/0143659042000185417.

Felt, U. 2018. “Responsible Research and Innovation.” In *Handbook of Genomics, Health and Society*, edited by S. Gibbon, B. Prainsack, S. Hilgartner, and J. Lamoreaux. London, New York: Routledge.
Finley, K. 2012. “The Military-Maker Complex: DARPA Infiltrates the Hackerspace Movement.” http://technoccult.net/archives/2012/02/24/the-militarymaker-complex-darpa-infiltrates-the-hackerspace-movement/.
Gianni, R., J. Pearson, and B. Reber. 2019. *Responsible Research and Innovation: From Concepts to Practices*. Abingdon, Oxon: Routledge.
Gibbons, M. 1999. “Science’s New Social Contract with Society.” *Nature* 402 (S6761): C81–C84. doi:10.1038/35011576.
Hertz, G. 2012. *Critical Making*. Hollywood, CA: Telharmonium. http://conceptlab.com/criticalmaking.
Hess, D. J. 2009. *Alternative Pathways in Science and Industry: Activism, Innovation and the Environment in an Era of Globalization*. Cambridge, MA: The MIT Press.
Irwin, A. 1995. *Citizen Science. A Study of People, Expertise and Sustainable Development*. Routledge.
Kieslinger, B., T. Schaefer, C. M. Fabian, E. Biasin, R. R. Freire, N. Mowoh, N. Arif, and P. Melis. 2021. “Covid-19 Response from Global Makers: The Careables Cases of Global Design and Local Production.” *Frontiers in Sociology* 6: 629587. doi:10.3389/fsoc.2021.629587.
Kohtala, C. 2016. *Making Sustainability: How Fablabs Address Environmental Issues (Doctoral D)*. Helsinki: Aalto University Publication.
Lange, B. 2017. “Offene Werkstätten Und Postwachstumsökonomien: Kollaborative Orte Als Wegbereiter Transformativer Wirtschaftsentwicklungen?” *Zeitschrift Für Wirtschaftsgeographie* 61 (1): 38–55. doi:10.1515/zfw-2016-0029.
Lindtner, S., and C. Lin. 2017. “Making and Its Promises.” *CoDesign* 13 (2): 70–82. doi:10.1080/15710882.2017.1308518.
Lindtner, S., S. Bardzell, and J. Bardzell. 2016. “Reconstituting the Utopian Vision of Making.” In *Conference on Human Factors in Computing Systems - Proceedings*, 1390–1402. Helsinki: Association for Computing Machinery. doi:10.1145/2858036.2858506.
Mamidipudi, A., and N. Frahm. 2020. “Turning Straw to Gold: Mobilising Symmetry in Responsible Research and Innovation.” *Science, Technology and Society* 25 (2): 223–239. doi:10.1177/0971721820902964.
Maxigas, P. 2012. “Hacklabs and Hackerspaces: Tracing Two Genealogies.” *Journal of Peer Production* 2. http://peerproduction.net/issues/issue-2/peer-reviewed-papers/hacklabs-and-hackerspaces/.
Nuesse, I., and R. Wanalo. 2019. “How Can Makerspaces Help Build Climate Change Resilience?” 2019. https://wellbeingeconomy.org/how-can-makerspaces-help-build-climate-change-resilience.
Ong, A., and S. J. Collier. 2004. *Global Assemblages: Technology, Politics, and Ethics as Anthropological Problems*. London: Wiley-Blackwell.
Paltrinieri, Roberta, and Piergiorgio Esposti. 2013. “Processes of Inclusion and Exclusion in the Sphere of Prosumerism.” *Future Internet* 5 (1): 21–33. http://www.mdpi.com/1999-5903/5/1/21.
Ratto, M. 2019. “Not Just Guns but Bullets, Too: ‘deconstructive’ and ‘Constructive’ Making Within the Digital Humanities.” In *Debates in the Digital Humanities Part III*, Chapter 26, edited by M. Gold and L. Klein, 307–318. Minneapolis, MN: University of Minnesota Press.
Ratto, M., and M. Boler. 2014. *DIY Citizenship. Critical Making and Social Media*. Cambridge: MIT Press.
Ratto, M., and G. Hertz. 2019. “Critical Making and Interdisciplinary Learning: Making as a Bridge Between Art, Science, Engineering and Social Interventions.” In *The Critical Makers Reader: (Un)Learning Technology Vol. #12. INC Reader*, edited by L. Bogers and L. Chiappini, 17–28. Amsterdam: Institute of Network Cultures, Amsterdam University of Applied Sciences.
Ratto, M., and S. Hockema. 2009. “FLWR PWR – Tending the Walled Garden.” In *Walled Garden*, edited by A. Dekker and A. Wolfsberger, 51–62. Amsterdam: Virtueel Platform.
Saari, H., M. Åkerman, B. Kieslinger, J. Myllyoja, and R. Sipos. 2021. “How Open Is the Maker Movement? Integrative Literature Review of the Openness Practices in the Global Maker Movement.” *Sustainability* 13 (24): 13559. doi:10.3390/su132413559.

Scholten, V. E., and V. Blok. 2015. “Foreword: Responsible Innovation in the Private Sector.” *Journal on Chain and Network Science* 15, SPECIAL ISSUE: Responsible innovation in the private sector. doi:10.3920/JCNS2015.x006.

Schumacher, E. F. 1973. *Small Is Beautiful: Economics as If People Mattered*. London: Blond & Briggs.

Selin, C., K. C. Rawlings, K. de Ridder-Vignone, J. Sadowski, C. A. Allende, G. Gano, S. R. Davies, and D. H. Guston. 2017. “Experiments in Engagement: Designing Public Engagement with Science and Technology for Capacity Building.” *Public Understanding of Science* 26 (6): 634–649. doi:10.1177/0963662515620970.

Sipos, R., M. Akerman, H. Saari, and B. Kieslinger. 2022a. Critical Making Responsibility Framework. Extending an Academic Proposal to Support Reflexivity in Maker Communities. doi:10.5281/zenodo.7432068.

Sipos, R., M. Åkerman, and V. Wenzelmann. 2022b. Critical Making Baseline. doi:10.5281/ZENODO.6907490.

Sipos, R., and V. Wenzelmann. 2021. *Critical Making With and For Communities: Community–Driven Critical Making Grounded in Practitioners’ Perspectives on Definition and Praxis*. C&T’21: proceedings of the 10th International Conference on Communities & Technologies – Wicked Problems in the Age of Tech, Seattle, USA. doi:10.1145/3461564.3461572.

Smith, A. 2005. “The Alternative Technology Movement: An Analysis of Its Framing and Negotiation of Technology Development.” *Human Ecology Review* 12 (2): 106–119. doi:10.22459/HER.26.01.2020.

Smith, A., J. M. Fressoli, D. Abrol, E. Arond, and A. Ely. 2017. *Grassroots Innovation Movements*. London: Routledge.

Stilgoe, J., S. J. Lock, and J. Wilsdon. 2014. “Why Should We Promote Public Engagement with Science?” *Public Understanding of Science* 23 (1): 4–15. doi:10.1177/0963662513518154.

Stilgoe, J., R. Owen, and P. Macnaghten. 2013. “Developing a Framework for Responsible Innovation.” *Research Policy* 42 (9): 1568–1580. doi:10.1016/j.respol.2013.05.008.

Tanenbaum, J. G., A. M. Williams, A. Desjardins, and K. Tanenbaum. 2013. “Democratizing Technology.” In *Conference on Human Factors in Computing Systems - Proceedings*, 2603–2612. doi:10.1145/2470654.2481360.

Tassone, V. C., C. O’Mahony, E. McKenna, H. J. Eppink, and A. E. J. Wals. 2018. “(Re-)Designing Higher Education Curricula in Times of Systemic Dysfunction: A Responsible Research and Innovation Perspective.” *Higher Education* 76 (2): 337–352. doi:10.1007/s10734-017-0211-4.

Unterfrauner, E., M. Hofer, B. Pelka, and M. Zirngiebl. 2020. “A New Player for Tackling Inequalities? Framing the Social Value and Impact of the Maker Movement.” *Social Inclusion* 8 (2): 190–200. doi:10.17645/si.v8i2.2590.

Von Schomberg, R. 2011. “Towards Responsible Research and Innovation in the Information and Communication Technologies and Security Technologies Fields.” *SSRN Electronic Journal*. doi:10.2139/ssrn.2436399.

Wylie, S. A., K. Jalbert, S. Dosemagen, and M. Ratto. 2014. “Institutions for Civic Technoscience: How Critical Making Is Transforming Environmental Research.” *The Information Society* 30 (2): 116–126. doi:10.1080/01972243.2014.875783.